

COTIVITI INTERN ASSESSMENT



Clinical Natural Language Technology for Health Care

Past, Present & Future Approaches

NLP · OCR · Computer Vision · LLM · LMM

What We'll Cover

01

Topic Definition

What is Clinical NLP and why does it matter?

02

Three Eras

Rule-based → Statistical/OCR → LLM/LMM

03

Market Trends

\$5.18B → \$16B by 2030 | Key drivers

04

Opportunities & Threats

Claims review, PHI compliance, hallucination risk

05

Strategic Recommendations

Hybrid CRR layer + human-in-the-loop scoring

06

Proof of Concept Demo

Live Streamlit hybrid extraction pipeline

What Is Clinical NLP?

Clinical NLP converts unstructured clinical text — physician notes, discharge summaries, claims narratives, and scanned records — into machine-readable, structured data that can be audited, analyzed, and applied in payment integrity decisions.

PAST

Rule-Based NLP

Regex patterns + clinical abbreviation dictionaries. Fast, deterministic, auditable.

PRESENT

Statistical / LLM

NER models, OCR, and LLMs reason across context and handle novel phrasing.

FUTURE

Multimodal AI

LMMs process text, images, and lab data together. Computer vision reads scanned forms.

Market Trends & Drivers

\$5.18B

NLP Market Size (2025)

MarketsandMarkets

\$16B

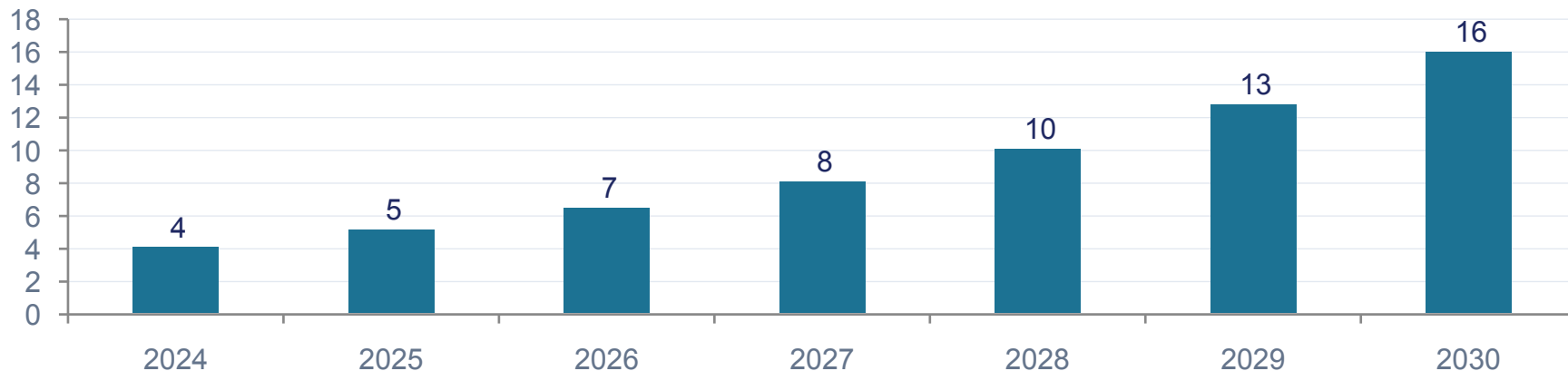
Projected by 2030

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~25%

CAGR Growth Rate

MarketsandMarkets



Opportunities & Threats

✓ Opportunities

Accelerated Claims Review

NLP validates medical necessity without reading full charts, increasing throughput.

Policy-to-Data Matching

Automated extraction surfaces mismatches between clinical notes and claim data.

Reduced Provider Abrasion

Fewer false positives on legitimate claims; less friction for providers.

⚠ Threats & Risks

LLM Hallucination

Rates up to 65.9% under default prompting; 44.2% with mitigation (PMC, 2026).

PHI Compliance (HIPAA)

Clinical text is dense with protected health info — the top barrier to deployment.

Detection Difficulty

Medical hallucinations use domain-specific language, hard for non-experts to catch.

Strategic Recommendations

for Cotiviti

01

Hybrid Clinical Record Review (CRR) Layer

Augment deterministic rule-based extraction with an LLM layer for novel or ambiguous phrasing. Rules handle well-defined patterns (fast, auditable); LLM handles the tail. Hallucination exposure is constrained because the LLM is only invoked where rules fail.

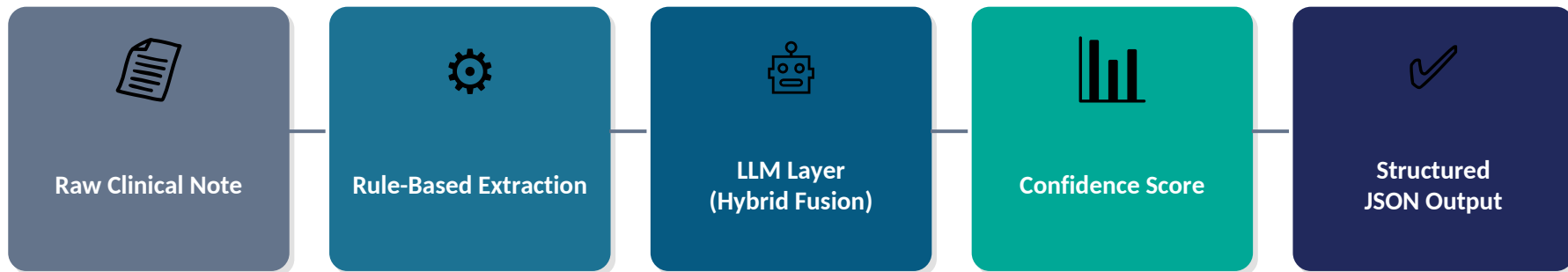
02

Human-in-the-Loop Confidence Scoring

Score confidence for each extraction and auto-route lower-confidence or higher-dollar-impact claims to human reviewers. Maximizes automation throughput without accepting the patient-safety and compliance risk of fully automated claim decisions.

Cotiviti's existing claims-validation model makes clinical NLP a natural extension — not an add-on.

Proof of Concept: Hybrid Clinical NLP Pipeline



What the POC Demonstrates:

- ▶ **clinical_extractor.py**: Pure rule-based NLP (regex + abbreviation dictionary). Zero dependencies, fully deterministic.
- ▶ **app.py (Streamlit)**: Hybrid pipeline: rule-based layer + simulated LLM fusion + confidence scoring with live UI.
- ▶ **First Principles**: Raw clinical text → diagnosis, medications, vitals, follow-up — structured JSON for downstream systems.
github.com/Rechal03/cotiviti-intern-assessment

Thank You

Clinical NLP is not an add-on for Cotiviti —
it's a natural extension of what you already do.

 Repository: github.com/Rechal03/cotiviti-intern-assessment

 Topic: Clinical NLP: Past, Present & Future